# Smart-Edu — Tutor Dashboard UI

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## Smart-Edu — Tutor Dashboard UI

### Complete Development Log

**Project:** Smart-Edu — Edu-Tech Platform

**Repository:** https://github.com/Lokesh-Sohanda8/Edu-Tech-Platform-SmartEdu

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# 1. Project Context

## 1.1 About Smart-Edu

Smart-Edu is a **full-stack EdTech tutoring management platform** built to help tutors and coaching institutes manage their entire teaching ecosystem.

The system allows tutors to manage:

- Students
- Courses
- Timetables
- Tests and marks
- Study materials
- Fees and payments
- Certificates
- Analytics
- Activities and schedules

Smart-Edu is structured as a **SaaS platform**.

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## 1.2 System Architecture

| Layer | Technology | Location |
|---|---|---|
| Frontend | Next.js 14 + Tailwind CSS | `client/` |
| Backend | Node.js + Express | `server/` |

| Layer | Technology | Location |
|---|---|---|
| Database | PostgreSQL | `PostgreSQL-DB/` |
| Shared Types | TypeScript Interfaces | `shared/` |
| Infrastructure | Docker | `docker/` |

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## 1.3 Purpose of Tutor Dashboard

The **Tutor Dashboard** is the primary workspace where tutors manage daily operations.

Key responsibilities of the dashboard:

- Monitor student activity
- Manage courses
- Track fee payments
- Schedule classes
- Upload study materials
- View analytics
- Track student performance

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# 2. Development Environment

## Technology Stack

| Category | Technology |
|---|---|
| Frontend Framework | Next.js 14 (App Router) |
| Language | TypeScript |
| Styling | Tailwind CSS |
| State | React State Hooks |
| Version Control | Git |
| Repository | GitHub |

| Category | Technology |
| --- | --- |
| Runtime | Node.js |

# 3. Initial Folder Structure

Initial structure created for the dashboard UI.

```
client/
├── src/
│
├── app/
│   ├── layout.tsx
│   ├── page.tsx
│
│   └── dashboard/
│       └── tutor/
│           ├── layout.tsx
│           ├── page.tsx
│
│           ├── students/
│           ├── courses/
│           ├── timetable/
│           ├── tests/
│           ├── materials/
│           ├── fees/
│           ├── enquiries/
│           ├── todo/
│           ├── activities/
│           ├── certificates/
│           └── analytics/
│
├── components/
│   ├── layout/
│   └── dashboards/
│
```

```
└── styles/
    └── globals.css
```

# 4. UI Architecture Design

The Tutor Dashboard follows a **modular UI architecture**.

Main layout structure:

```
Dashboard Layout
│
├── Sidebar
│
├── Top Header
│
└── Main Content Area
      │
      ├── Dashboard Widgets
      ├── Data Tables
      ├── Cards
      └── Analytics Panels
```

# 5. Component Creation

Major reusable components were created for modular UI design.

## Sidebar Component

### Location

```
src/components/layout/Sidebar.tsx
```

### Purpose

Provides navigation between dashboard modules.

Features:

- Navigation items
- Active route detection
- Collapsible sidebar

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## Top Header Component

**Location**

```
src/components/layout/TopHeader.tsx
```

**Purpose**

Global dashboard header containing:

- Search bar
- Notifications
- Profile menu
- Logout option

---

## Dashboard Layout Component

**Location**

```
src/components/layout/DashboardLayout.tsx
```

This layout wraps all tutor dashboard pages.

Structure:

```
Sidebar
   |
TopHeader
```

```
    |
Children Pages
```

# 6. Pages Created

The Tutor Dashboard consists of **11 main pages**.

| Page | Route |
|---|---|
| Dashboard Home | `/dashboard/tutor` |
| Students | `/dashboard/tutor/students` |
| Courses | `/dashboard/tutor/courses` |
| Timetable | `/dashboard/tutor/timetable` |
| Tests | `/dashboard/tutor/tests` |
| Materials | `/dashboard/tutor/materials` |
| Fees | `/dashboard/tutor/fees` |
| Enquiries | `/dashboard/tutor/enquiries` |
| Todo Tasks | `/dashboard/tutor/todo` |
| Activities | `/dashboard/tutor/activities` |
| Certificates | `/dashboard/tutor/certificates` |
| Analytics | `/dashboard/tutor/analytics` |

# 7. Routing System

Smart-Edu uses **Next.js App Router**.

Folder-based routing example:

```
src/app/dashboard/tutor/students/page.tsx
```

Route generated automatically:

```
/dashboard/tutor/students
```

## Layout Persistence

Next.js `layout.tsx` ensures persistent UI elements.

```
Sidebar
TopHeader
```

These are rendered once and persist across navigation.

Only the **page content changes**.

## Active Route Detection

Sidebar highlights the current page.

Example code:

```
const pathname = usePathname();
const isActive = pathname === item.href;
```

# 8. UI Design System

## Color Palette

| Token | Color | Usage |
|---|---|---|
| primary | #4F6EF7 | Primary buttons and links |
| success | #22C55E | Success states |
| warning | #F97316 | Alerts |
| danger | #EF4444 | Errors |
| sidebar | #0F172A | Sidebar background |
| surface | #F8F9FC | Page background |

## Typography

Font: **Inter**

| Size | Usage |
|---|---|
| 12px | Labels |
| 14px | Body text |
| 16px | Card titles |
| 20px | Page titles |
| 24px | Stat numbers |

## Spacing System

Tailwind spacing scale used.

| Pattern | Value |
|---|---|
| Cards | `p-5` or `p-6` |
| Grid gaps | `gap-4` / `gap-6` |
| Sections | `space-y-6` |

# 9. Errors Encountered During Development

## Error 1

### Next.js Config File Not Supported

Error:

```
Configuring Next.js via next.config.ts is not supported
```

Cause:

Next.js 14 does not support TypeScript config files.

Fix:

Created new config:

```
const nextConfig = {};
module.exports = nextConfig;
```

Deleted:

```
next.config.ts
```

---

# Error 2

## GitHub File Size Limit

Error:

```
File exceeds GitHub limit of 100MB
```

Cause:

`node_modules` was committed.

Fix:

```
git rm -r --cached client/node_modules
```

Updated `.gitignore`.

---

# Error 3

## Wrong GitHub Repository

Push was going to the wrong repository.

Fix:

```
git remote set-url origin
```

Correct repository was set.

---

## Error 4

### Git History Divergence

Error:

```
non-fast-forward push rejected
```

Fix:

```
git push origin main --force
```

---

# 10. Git Operations Performed

### Files Added

- Next.js config files
- Tailwind config
- Layout components
- Dashboard pages
- Global styles

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### Git Commands Used

```
git add -A
git commit -m "Add SmartEdu Tutor Dashboard UI"
git remote set-url origin <repo>
git push origin main --force
```

---

### .gitignore

```
node_modules/
.next/
.env
*.tsbuildinfo
next-env.d.ts
```

---

# 11. Final Folder Structure

```
client/
│
├── package.json
├── tsconfig.json
├── next.config.js
├── tailwind.config.ts
│
└── src/
    ├── styles/
    │   └── globals.css
    │
    ├── app/
    │   ├── layout.tsx
    │   ├── page.tsx
    │   │
    │   └── dashboard/
    │       └── tutor/
    │
    │           ├── layout.tsx
    │           ├── page.tsx
    │
    │           ├── students/
    │           ├── courses/
    │           ├── timetable/
    │           ├── tests/
```

```
    │           ├── materials/
    │           ├── fees/
    │           ├── enquiries/
    │           ├── todo/
    │           ├── activities/
    │           ├── certificates/
    │           └── analytics/
    │
    └── components/
        ├── layout/
        │   ├── Sidebar.tsx
        │   ├── TopHeader.tsx
        │   └── DashboardLayout.tsx
        │
        └── dashboards/
            ├── StatCards.tsx
            ├── CalendarWidget.tsx
            ├── AlertsPanel.tsx
            ├── AIInsightsPanel.tsx
            └── ActivityFeed.tsx
```

---

# 12. UI Screens Overview

## Dashboard Home

Displays:

- 4 Stat Cards
- Alerts Panel
- Activity Feed
- Calendar Widget
- AI Insights Panel

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## Students Page

Features:

- Student table
- Search functionality
- Status badges
- Pagination

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## Courses Page

Shows:

- Course cards
- Progress bars
- Student counts
- Create new course option

---

## Timetable Page

Weekly grid schedule:

- Subject blocks
- Time slots
- Color coded classes

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## Materials Page

File management interface:

- Upload area
- File list
- Download / Delete actions

---

## Fees Page

Financial dashboard:

- Fee summary
- Payment table
- Status badges

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# 13. Future Improvements

## Backend Integration

Replace mock data with API endpoints:

```
GET /api/tutor/students
GET /api/tutor/courses
POST /api/tutor/fees
```

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## Authentication

Add:

- JWT authentication
- Route protection
- Tutor session management

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## Real-Time Notifications

Possible technologies:

- WebSockets
- Server-Sent Events

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## AI Analytics

Future ML features:

- Score prediction
- Dropout risk detection
- Performance analytics

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## Mobile Optimization

Add:

- Responsive sidebar
- Mobile navigation
- PWA support

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